

The Price of Thinking: Reasoning Effort as a Model-Specific API Contract

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Abstract

API buyers purchase a dated contract, not a model name alone: the contract includes the requested and served model, reasoning-effort term or its omission, output rail, service product, prompt, and price schedule. We study the reasoning-effort term through a registered paired contrast of Sonnet 5 with explicit high effort against the same model with effort omitted, using 30 AIME 2026 items and five calls per item. Every paid attempt was assigned one frozen terminal category, and inference resampled items while retaining their repeated calls. Mean delivered cost was \$0.01031 per call higher under the explicit-high contract than under the omitted contract [$+\$0.00204$, $+\$0.01974$]. The corresponding accuracy contrast was $+0.0133$ [-0.0267 , $+0.0467$]; we did not detect an accuracy difference, and the interval permits a gain of up to 4.67 percentage points that this design cannot rule out. Cost per correct answer was \$0.08665 under the high-effort contract and \$0.07662 under the omitted contract, as registered point estimates. A dated contract census, Models-API metadata, and preregistered raw-response probes further documented model-specific omission semantics, including within a provider; claims remained at documentation grade when raw structure was indeterminate. The request registry, parser, terminal taxonomy, statistical plan, and analysis pipeline were frozen before outcomes were examined; the resulting claims are bounded to the model, task, and collection date studied.

Introduction

The price printed beside a language model is not the price of a completed task. API inference cost was already stochastic when buyers paid for variable-length completions, but reasoning models add a less legible source of variation: hidden thinking tokens allocated before the visible answer. Chen et al. (2026) report that realized cost reversed the ordering implied by listed token prices in roughly one in three of 336 model-task comparisons and that repeated calls to fixed queries varied substantially in cost. We take those results as established background, not as contributions of this study.

That evidence leaves a narrower question unresolved. Chen et al. (2026) run each model at one reasoning setting and identify the effort-cost ablation as future work. Their unit of comparison is therefore a configured model: changes in provider, architecture, token prices, and reasoning mode can move together. They also analyze cost separately from quality. Incorrect responses retain their token costs, but paid attempts are not classified into mutually exclusive delivered outcomes that

distinguish correct answers, wrong answers, rail-exhausted no-answer responses, other no-answer responses, and provider failures. Those choices fit a study of model-level price reversals, yet they do not identify what a buyer receives by changing an effort parameter while keeping the model fixed. Nor do they show whether two request surfaces that expose the same nominal control implement the same omission behavior. The relevant object for that question is not the provider or model name alone, but the contract formed by the requested and served model, effort setting or omission, output rail, service product, prompt, price schedule, and run date.

We ask: given that reasoning-model cost is stochastic, what changes when a buyer changes one explicit term of the same model’s API contract, and how should the resulting delivered outcomes be priced? Our primary design is a registered, paired contrast of Sonnet 5 with explicit high effort against Sonnet 5 with the effort parameter omitted. Effort and thinking are separate request fields. In the omitted cell both were unspecified: Sonnet 5’s documented defaults were high effort and adaptive thinking. Disabling thinking requires an explicit `thinking: {"type": "disabled"}` request (Exhibit 1). The primary cells therefore differ in whether high effort is explicit or defaulted, not in their documented effort level or thinking mode. The comparison holds the model, prompt template, 30 AIME 2026 items, output rail, service product, and price schedule fixed; each item receives five calls under each contract. The full grid adds single-pass reference cells for GPT-5.6 Terra with effort omitted and Claude Fable 5 with effort omitted, plus a small non-contemporaneous GPT-5.4-mini low- and high-effort bridge. Every attempted purchase receives both a delivered-cost measurement and one terminal category under a frozen precedence rule. A valid final answer is correct or wrong regardless of stop state; a rail outcome requires both rail exhaustion and no valid final answer; other no-answer and provider-failure states remain separate. We estimate cell means and paired contrasts by resampling items while retaining all calls for each selected item. Repeated calls therefore reveal conditional behavior on a problem without being counted as additional independent problems.

Mean delivered cost was \$0.01031 per call higher under the explicit-high contract than under the omitted contract. The point estimate of the corresponding accuracy contrast was +0.0133, but we did not detect an accuracy difference; the registered interval permits a gain of up to 4.67 percentage points that some buyers could value. The result must not be read as evidence that high effort buys nothing. The registered cost-per-correct point estimate was also higher under the high-effort contract than under the omitted contract, but no separate interval was registered. Three supporting contributions place that contrast in an auditable market setting. First, a dated contract census and preregistered raw-response probes document that omission semantics differ across models, including within one provider, so provider-level descriptions are too coarse. Raw structure verified some exact-model claims, while structurally indeterminate responses left others at documentation grade. Second, a frozen terminal-outcome taxonomy keeps wrong and no-answer calls in the purchased outcome mixture and its cost total instead of treating them as missing observations. Third, the request registry, spending limits, parser, statistical rules, and analysis pipeline were fixed before the registered outcomes were examined. Together these elements turn an effort label into a measurable contract comparison while keeping the empirical claim bounded to the model, task, and collection date studied here.

Related work

Cost-aware deployment research treats model selection as an economic decision rather than a pure accuracy ranking. Chen, Zaharia, and Zou (2023) formalize this decision in FrugalGPT as

cost-constrained utility maximization and learn cascades that route a query through a ladder of models, stopping when a cheaper response is adequate and escalating otherwise. In that formulation, the available services and their cost information are inputs to the routing problem: the router chooses among models whose per-query prices are treated as known inputs. Our setting changes the object being chosen, not the logic of cost-aware routing. A buyer can alter a reasoning-effort term within one model, and omission is itself a model-specific request state. We therefore treat the configured API contract as the candidate service. We do not claim to have discovered a mismatch between listed and realized prices; that finding belongs to Chen et al. (2026), discussed below.

Cost-adjusted evaluation asks how capability comparisons change once inference expense enters the objective. Erol et al. (2025) define Cost-of-Pass, an economically framed frontier metric that combines task success with inference cost across models and strategies. Our cost-per-correct quantity belongs to the same family: total delivered cost is divided by the number of correct outcomes. We do not propose it as a new metric. Instead, we apply the quantity within one commercial model across two registered effort contracts, report it as a point estimate, and retain every paid attempt in the cost and terminal-outcome accounting. Cost-of-Pass supplies the decision framing; the present study asks whether changing a request term changes the purchased outcome under an otherwise fixed model contract rather than constructing another cross-model frontier.

The FrugalGPT authors themselves return to this problem in *The Price Reversal Phenomenon* (Chen et al., 2026), which examines reasoning-model costs at substantially greater model and task breadth. They report that listed token prices frequently misorder realized workload costs — roughly one in three of 336 comparisons in the pinned v2 — and attribute the reversals to thinking-token volume on single-turn tasks and to additional turn and context effects in agentic settings. They also report substantial cost variation across repeated calls to fixed queries. We take these as established results, not contributions of this paper. Their design runs each model at a single reasoning setting, analyzes cost separately from quality, and does not classify paid attempts by terminal outcome. Our narrower point of departure is the registered within-model contrast between explicit high effort and the omitted contract, coupled to item-clustered inference and joint accounting of delivered cost and terminal outcomes.

Methods

Task and items

The accuracy instrument is the complete 30-item AIME 2026 cohort distributed through the MathArena evaluation platform (Dekoninck et al., 2026) and the pinned `MathArena/aime_2026` revision. We preserve the ordered measurement IDs `aime_2026_i_01` through `aime_2026_i_30`. A model response is generated freely, but the scoring target is closed-form: the final claimed answer must be an integer from 0 through 999. The frozen parser reads the final explicit **Answer:** N claim and does not fall back to an earlier valid-looking claim when the final claim is malformed.

AIME 2026 was chosen to reduce the contamination concern observed on the prior year’s problems. The contest was administered in February 2026, after the documented knowledge cutoffs used in the registered design, although the gap from Anthropic’s January 2026 cutoff was narrow. In the preliminary cohort, Sonnet low answered two AIME 2025 items correctly in about 10 billed output tokens each, too few to contain hidden reasoning. At the same item indices in AIME 2026, billed output rose to 755 and 1,627 tokens while the visible reply remained a short final claim, indicating substantial hidden computation. We treat this within-index change as a retrieval-like

freshness diagnostic, not as proof that AIME 2026 is uncontaminated. AIME material is licensed CC BY-NC-SA. The release therefore contains item identifiers, configurations, measurements, and derived outcomes, but not the problem text or reference answers.

Contracts and registered cells

The unit of comparison is a contract cell: requested model, served model, effort term or its omission, output rail, service product, request mode, prompt template, dated price schedule, and collection date. This definition prevents a model-specific result from being promoted to a provider-wide claim and makes a served-model or service-tier mismatch an observable contract deviation rather than an ignorable implementation detail.

The main grid contains four registered cells on all 30 items. We refer to the full contracts as Sonnet high, Sonnet omitted, Terra omitted, and Fable omitted. The two Sonnet cells each use five calls per item; Terra omitted and Fable omitted each use one. Repeat depth was concentrated in the paired Sonnet contrast, where conditional within-item behavior is a registered estimand; the single-pass cells serve as cross-model reference points. Two earlier `gpt-5.4-mini` cells supply a five-item descriptive bridge. Every cell uses the frozen `aime_frozen_v1` prompt template and a 64,000-token output rail.

Table 1. Registered AIME 2026 contract cells and descriptive bridge cells.

Cell	Requested model and effort term	Items x calls per item	Calls	Collection
Sonnet high	<code>claude-sonnet-5</code> , explicit high	30 x 5	150	2026-07-18
Sonnet omitted	<code>claude-sonnet-5</code> , effort omitted	30 x 5	150	2026-07-18
Terra omitted	<code>gpt-5.6-terra</code> , effort omitted	30 x 1	30	2026-07-18
Fable omitted	<code>claude-fable-5</code> , effort omitted	30 x 1	30	2026-07-18
Mini low bridge	<code>gpt-5.4-mini</code> , explicit low	5 x 1	5	2026-07-09
Mini high bridge	<code>gpt-5.4-mini</code> , explicit high	5 x 1	5	2026-07-09

The 360 main-grid calls were made in the single registered session `aime2026_main_20260718_01` using non-streaming real-time requests and no automatic retries. Each call was issued as a separate, stateless API request: no conversation history or earlier response was supplied, and the study passed no state from one call to the next. Repeats therefore sampled the same frozen request contract rather than a conversation in which the model had already seen the item. Dispatch order was fixed rather than randomized: within all 150 Sonnet item/repeat pairs, the high call preceded the omitted call in the queue. The first 20 pairs ran strictly sequentially; the remaining 130 used bounded concurrency, with at most three in-flight requests per provider, under which completion order could differ. Every successful main-grid response reported the exact requested model ID. The bridge alias resolved to `gpt-5.4-mini-2026-03-17`; its five shared items were indices 1, 5, 8, 12, and 15. Because those bridge calls are non-contemporaneous and cover only five items, their comparisons remain descriptive.

Dated contract census and structural probes

Before main-grid promotion, we froze a July 14 census of seven current-menu model IDs using official documentation and Models-API metadata, without inference calls. On July 16, we then ran 19 preregistered, single-item contract probes on the preselected HEADLINES task. These probes tested request acceptance, served identity, echoed effort, thinking-block structure, thinking-token details, and explicit disabled-thinking requests; their task accuracy was not used to select main-grid cells.

Evidence grades followed a fixed hierarchy. A raw echoed effort value, positive thinking structure, or acceptance or rejection of an explicit disabled-thinking request upgraded only the contract property it directly identified. By contrast, an absent thinking block together with a missing or zero thinking-token field could not distinguish disabled thinking from adaptive thinking that selected zero tokens. Such a result remained indeterminate and did not upgrade documented omission behavior.

Delivered cost and terminal outcomes

Delivered cost is reconstructed from provider-reported usage under the price schedule registered for each cell. The reconstruction uses the provider’s applicable input, cached-input, and output accounting, including billed reasoning within output usage, and is checked against the cost stored with each row. For the July 18 main session, provider-level reconstructed totals were also reconciled against authenticated console balance changes at the console’s cent granularity. This external reconciliation validates the accounting boundary without changing any row or statistical estimate.

Every attempted call receives exactly one of five terminal categories: `correct`, `wrong`, `no_answer_rail`, `no_answer_other`, or `provider_failure`. Classification follows a fixed precedence. A valid final claim is correct or wrong regardless of the provider stop state. A call is `no_answer_rail` only when it exhausts its registered output rail and contains no valid final claim; a successful response without a valid claim and without rail exhaustion is `no_answer_other`. Transport, timeout, rate-limit, server, and rejected-request failures enter `provider_failure` under the registered operational policy rather than being silently converted to wrong answers.

Raw provider responses are preserved before extraction. The analysis decodes those artifacts, extracts visible text again, applies the frozen parser, and combines the fresh parse with the normalized provider stop state. Consequently, a local extraction exception cannot turn a preserved, billed response into a provider failure. Wrong and no-answer calls remain in the attempt count, terminal-outcome mixture, and total delivered cost because they are purchased outcomes rather than missing observations.

Statistical plan

The task item is the independent unit. For a repeated cell, calls are first averaged within item; cell means then weight the 30 item-level values equally. Paired contrasts are formed within shared items before averaging. The primary uncertainty procedure is an item-clustered percentile bootstrap: each draw samples 30 items with replacement and carries every call and compared cell belonging to a selected item. We use 10,000 draws, analysis seed 20260713, and percentile 95% intervals. Repeats therefore estimate conditional behavior on a problem but never convert 30 items into 150 independent problems.

The nominal 95% level describes repeated use of the procedure, to the extent that the bootstrap approximation holds with 30 independent items; it is not a posterior probability that a realized

interval contains the estimand. At this sample size, percentile coverage can be imperfect or discrete, especially for sparse outcomes or rare extreme costs absent from the observed cohort. The bootstrap redistributes observed item clusters but cannot create an event that no observed item contains. Counts and item denominators therefore accompany sparse outcomes, and a zero-event interval that collapses to zero is not interpreted as an informative population upper bound.

For unrepeatable binary proportions, we additionally report two-sided exact 95% Clopper-Pearson intervals. This rule applies to the Terra and Fable main cells and the two mini bridge cells; repeated-call proportions are not treated as independent binomial trials. Sparse outcomes are accompanied by their raw counts and item denominators.

Tail reporting was gated before collection by the number of independent items. Median, p90, p95, and maximum are eligible with at least 20 items. A p99 requires at least 100 items, and 90%-CVaR requires at least 50. The 30-item main cohort is therefore eligible for the first set but not for p99 or CVaR. We do not report those ineligible quantities, even though software could calculate sample analogues, because the refusal was part of the registered analysis rather than a reaction to the observed tail.

In preliminary collection, repeated Sonnet high calls on one AIME item mixed cheap wrong answers with rail-censored no-answer responses, motivating the registered rail-utilization measures. Before main collection, we fixed the following preregistered conservative gate: if rail outcomes were absent from the main grid, the tail claim would narrow to the preliminary existence evidence and registered upper bounds; if they recurred, we would describe their observed frequency without extrapolating beyond supported quantiles.

Governance and reproducibility

The prompt, parser, terminal taxonomy, statistical rules, contract registry, spending ceilings, and executable analysis pipeline were frozen before any registered outcome was inspected. Raw session and analysis artifacts are immutable and SHA-256-pinned; the integrity validator fails on missing, extra, truncated, or changed registered files. Code development for the final analysis used synthetic fixtures, and the first successful production invocation was the first read of the main-session correctness fields. Any later deviation must be recorded as a dated amendment stating whether the affected outcomes had already been observed; unregistered statistics do not enter the confirmatory results.

Results

Primary Sonnet contract contrast

Mean delivered cost was \$0.01031 per call higher under the explicit-high contract than under the omitted contract [$+\$0.00204$, $+\$0.01974$], while the accuracy contrast was $+0.0133$ [-0.0267 , $+0.0467$]. Cost per correct answer was \$0.08665 under Sonnet high and \$0.07662 under Sonnet omitted; these are point estimates without separately registered intervals (Table 2).

Table 2. Accuracy, delivered cost, and observed cost tails by registered AIME 2026 contract.

Contract	Calls (items x repeats)	Accuracy (correct/calls), 95% CI	Mean cost/call, 95% CI	Cost/correct	Cost tail: median / p95 / max
GPT-5.6 Terra, effort omitted	30 (30 x 1)	0.933 (28/30) [0.779, 0.992]	\$0.02319 [\$0.01745, \$0.02992]	\$0.02485	\$0.01801 / \$0.05093 / \$0.08884
Claude Sonnet 5, effort omitted	150 (30 x 5)	0.913 (137/150) [0.813, 0.987]	\$0.06998 [\$0.04427, \$0.10062]	\$0.07662	\$0.04093 / \$0.24428 / \$0.53290
Claude Sonnet 5, explicit high effort	150 (30 x 5)	0.927 (139/150) [0.833, 1.000]	\$0.08029 [\$0.05107, \$0.11587]	\$0.08665	\$0.04169 / \$0.31741 / \$0.58653
Claude Fable 5, effort omitted	30 (30 x 1)	1.000 (30/30) [0.884, 1.000]	\$0.21582 [\$0.11158, \$0.36711]	\$0.21582	\$0.10466 / \$0.72662 / \$1.98876
GPT-5.4-mini, low (July 9 bridge; descriptive)	5 (5 x 1)	0.800 (4/5) [0.284, 0.995]	\$0.01006 [\$0.00401, \$0.01657]	\$0.01257	Not reported (5 items)
GPT-5.4-mini, high (July 9 bridge; descriptive)	5 (5 x 1)	0.800 (4/5) [0.284, 0.995]	\$0.04082 [\$0.00511, \$0.10396]	\$0.05102	Not reported (5 items)

Notes: Accuracy is shown as the observed call proportion with the raw correct/call count. Accuracy intervals for the repeated Sonnet cells use the registered item-clustered bootstrap, which resamples items while retaining all five calls; exact Clopper-Pearson intervals are shown for the single-pass and bridge cells. Repeated calls do not increase the independent-item count. Mean-cost intervals use the registered item-clustered bootstrap. Cost per correct is the frozen total delivered cost divided by the observed correct count and is copied from the registered statistics rather than recomputed for this table. Cost tails report the registered call-level median, p95, and maximum only for cells meeting the 20-independent-item eligibility gate. Bridge cells are descriptive, non-contemporaneous anchors and do not meet that gate. Dollar values use the dated price schedules registered for each contract and Python fixed-point formatting applied to the stored floating-point values. Latency is not represented.

Dated omission and thinking-control evidence

Exhibit 1. Contract-control evidence frozen before the main-grid outcomes were examined.

- **GPT-5.6 Sol, Terra, and Luna.** Documentation assigned omitted effort to **medium**. Each omitted probe echoed **reasoning.effort=medium**, raw-verifying that behavior for these exact model IDs and the probe date (**structural_raw_response**).
- **Claude Fable 5.** Documentation assigned omitted effort to **high** and described adaptive thinking as always on. The omitted probe contained positive thinking structure, while an explicit disabled-thinking request returned an HTTP 400 error naming the unsupported mode. These raw artifacts verify realized thinking on the omitted probe and the absence of an off-switch, not a general thinking-token level (**structural_raw_response**; **structural_request_error**).
- **Claude Opus 4.8.** Documentation assigned omitted effort to **high** but described omitted thinking as off. The omitted probe contained no thinking block and reported zero thinking tokens, which was consistent with but did not raw-verify that behavior. A disabled-thinking-plus-high request was accepted. Omission therefore remained documentation-grade

with indeterminate raw evidence (`documentation_plus_indeterminate_raw`; `request_surface_raw_response` for the accepted disabled request).

- **Claude Sonnet 5.** Documentation assigned omitted effort to `high` and described omitted thinking as adaptive. The omitted probe contained no thinking block and reported zero thinking tokens, which could not distinguish adaptive-zero from disabled thinking. A disabled-thinking-plus-high request was accepted. Omission therefore remained documentation-grade with indeterminate raw evidence (`documentation_plus_indeterminate_raw`; `request_surface_raw_response` for the accepted disabled request).
- **Claude Haiku 4.5.** Documentation and Models-API metadata exposed no effort control. The probe served the pinned model snapshot without thinking structure, providing consistency evidence but not testing an unrequested effort parameter (`metadata_plus_structural_raw_response`).

The exhibit separates documented behavior from structural raw-response evidence rather than treating every successful probe as verification. The dated source record is preserved in the dated census, contract registry, probe expectation, and probe result records.

Descriptively, the registered grid rows bear on the same question: 140 of the 150 Sonnet omitted calls delivered positive reasoning-token counts (explicit high: 141 of 150), with positive-call medians of 3,133.5 versus 3,359 tokens. Positive counts show that thinking occurred in those 140 omitted calls and therefore that omission did not act as disabled thinking in them. The two medians are post hoc descriptive quantities and do not establish equivalence between request forms or a causal effect of explicitly setting high. None of these values is a registered statistic.

Delivered-cost distributions

Figure 1 displays all 360 registered calls by contract and terminal category. Median delivered cost was \$0.01801 for Terra omitted, \$0.04093 for Sonnet omitted, \$0.04169 for Sonnet high, and \$0.10466 for Fable omitted. The two Sonnet contracts had similar medians even though their individual calls occupied visibly different parts of the cost range. Both also contained a low-cost cluster that included unsuccessful outcomes; we treat that cluster qualitatively rather than imposing an unregistered cost threshold or reporting a post hoc share. Each dot in the figure remains a call nested within an item. The display therefore exposes call-level overlap without treating the dots as independent evidence.

Accuracy-cost frontier and historical bridge

Figure 2 places the four main cells on the registered accuracy-cost plane. Terra omitted answered 28/30 items correctly, with exact interval [0.779, 0.992], at mean delivered cost \$0.02319 [\$0.01745, \$0.02992]. Sonnet omitted had accuracy 0.913 [0.813, 0.987] and mean cost \$0.06998 [\$0.04427, \$0.10062]; Sonnet high had accuracy 0.927 [0.833, 1.000] and mean cost \$0.08029 [\$0.05107, \$0.11587]. For Fable omitted, no failure was observed in 30 items; its exact accuracy interval was [0.884, 1.000], and mean cost was \$0.21582 [\$0.11158, \$0.36711]. Because all 30 calls were correct, its cost-per-correct point estimate was numerically identical to its mean cost.

The two July 9 mini bridge cells are descriptive and non-contemporaneous. Low and high each answered 4/5 items correctly, with exact interval [0.284, 0.995]; their mean costs were \$0.01006 [\$0.00401, \$0.01657] and \$0.04082 [\$0.00511, \$0.10396], respectively.

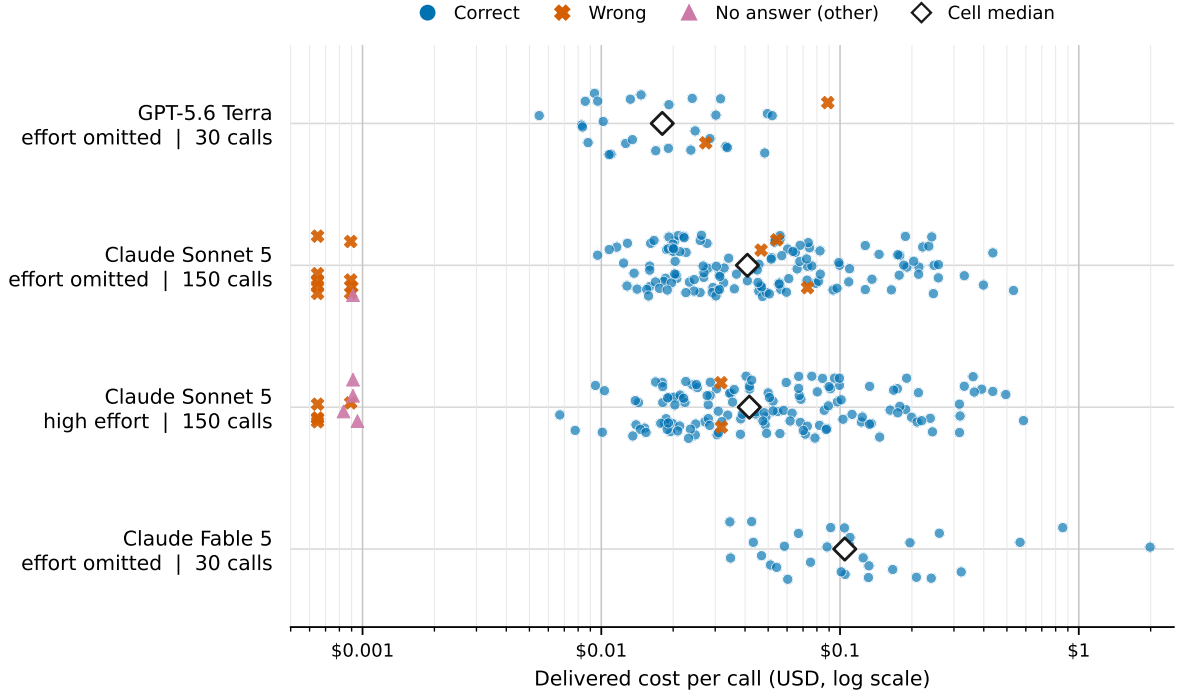


Figure 1. Delivered cost distributions under four registered AIME 2026 contracts. Each point represents one provider call. The two Sonnet cells contain five repeated calls on each of 30 items; Terra and Fable contain one call per item. The task item is the inferential unit, and repeated calls remain grouped within item rather than being treated as additional independent observations. The figure itself contains no uncertainty intervals. Color and shape indicate the terminal outcome, and open diamonds mark the registered cell medians. Delivered cost is shown on a logarithmic scale. Latency is not represented.

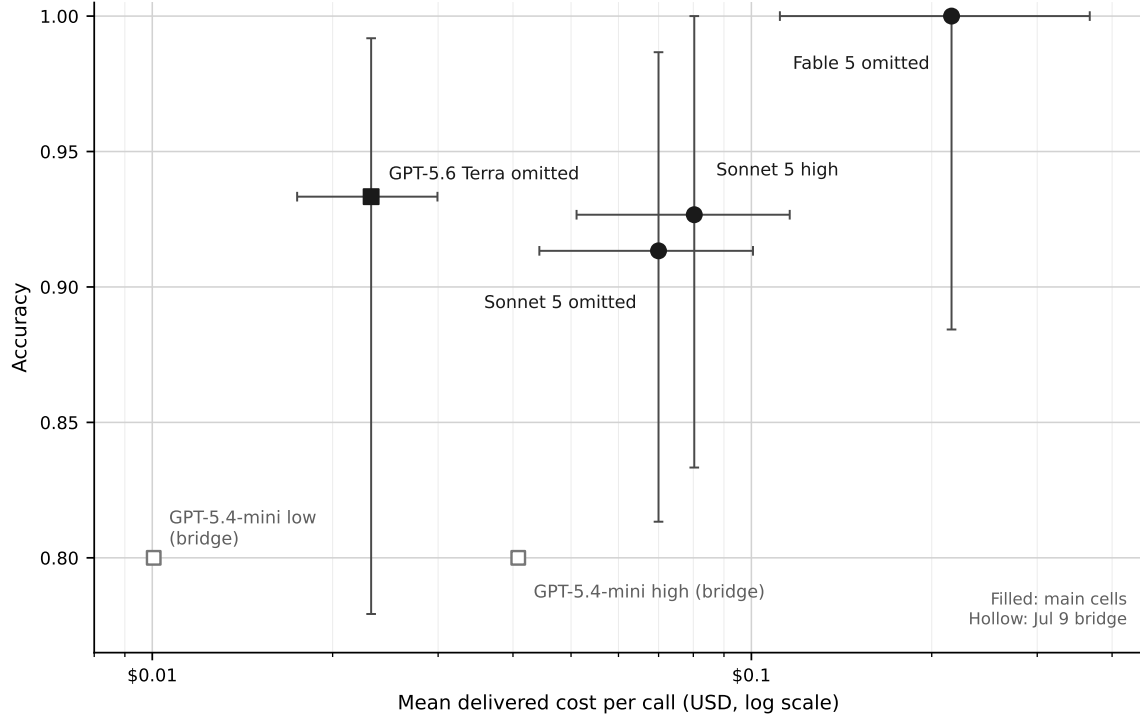


Figure 2. Accuracy and mean delivered cost under registered AIME 2026 contracts. Points locate mean delivered cost per registered call against accuracy. For the two repeated Sonnet cells, vertical intervals are registered 95% item-clustered bootstrap intervals. For the single-pass Terra and Fable cells, vertical intervals are two-sided 95% exact Clopper-Pearson intervals. Horizontal intervals are registered 95% item-clustered bootstrap intervals for mean delivered cost. The two smaller hollow points are the non-contemporaneous five-item GPT-5.4-mini bridge cells collected on July 9 and are shown without interval bars; each observed 4/5 correct, with exact 95% interval [0.284, 0.995]. Fable produced 30/30 observed correct responses, with exact 95% interval [0.884, 1.000]; this is not evidence of population accuracy equal to 100%. Mean delivered cost is shown on a logarithmic scale. Latency is not represented.

Within-item variation

Figure 3 compares the registered minimum, mean, and maximum delivered cost for each item in the repeated Sonnet cells; these spans are descriptive, not uncertainty intervals. Sonnet high produced more than one terminal category on items 2, 11, and 15, and Sonnet omitted did so on items 15, 17, 18, and 25. Item 11 was selected after outcome inspection as a descriptive illustration; the figure displays all 30 items, and no population-rate claim is based on that selection. At the stored six-decimal precision, Sonnet high on item 11 combined wrong and correct outcomes across a cost span from \$0.000648 to \$0.391888; under Sonnet omitted, all five calls were wrong and delivered cost was degenerate at \$0.000648. The preliminary item 15 mixture paired cheap wrong answers with rail-censored no-answers (Methods), a different second branch from item 11’s pairing of cheap wrong answers with one costly correct computation, so item 11 is not a replication.

Rail utilization and preregistered conservative gate resolution

No main-grid call ended in `no_answer_rail`, and none of the 360 calls reached 99% rail utilization. For Sonnet high, 7/150 calls reached at least 50% utilization, with item-bootstrap interval [0.0067, 0.1000]; 1/150 reached at least 90%, with interval [0.0000, 0.0200]; and 0/150 reached 99%. Sonnet omitted had 4/150 calls at 50% [0.0000, 0.0733] and 0/150 at both 90% and 99%. For these three zero-event repeated-cell estimates, all registered bootstrap draws were zero, mechanically yielding [0.0000, 0.0000]. They are not informative population upper bounds.

Fable omitted had 1/30 calls at 50%, with exact interval [0.0008, 0.1722], and 0/30 at both higher thresholds, each with exact interval [0.0000, 0.1157]. Terra omitted had 0/30 calls at every threshold, also with exact interval [0.0000, 0.1157] at each. The preliminary full-rail mixture therefore did not generalize, resolving the preregistered conservative gate (Methods). As specified in Methods, p99 and CVaR remain unreported. Rail utilization measures delivered output at the registered boundary and does not identify the unconstrained resources a censored call would have required.

Latency in the first-20-item sequential subsample (descriptive)

Latency results apply only to the registered first-20-item sequential subsample and are descriptive. Median latency was 7.8 s under Terra omitted, 17.1 s under Sonnet omitted, 16.7 s under Sonnet high, and 17.8 s under Fable omitted. Fable’s eligible upper-tail summaries were p95 226.1 s and maximum 482.9 s. These values describe the registered subsample and are not interpreted as causal effects of effort. They exclude the later bounded-concurrency phase, and the same scope restriction will accompany any latency table or figure.

Discussion

The primary result is a within-model contract contrast, not a provider ranking. The registered comparison detected higher delivered cost under the explicit-high contract but did not detect an accuracy difference. This result must not be read as showing that high effort buys nothing: the accuracy interval permits a gain of up to 4.67 percentage points, which a buyer could value at the observed premium. The registered cost-per-correct point estimate was also higher under Sonnet high, but no separate interval was registered. The narrower implication is that Sonnet high and Sonnet omitted are distinct request contracts within one model. This extends cost-aware routing’s model-level price input with a controlled contract choice; it does not claim to discover stochastic

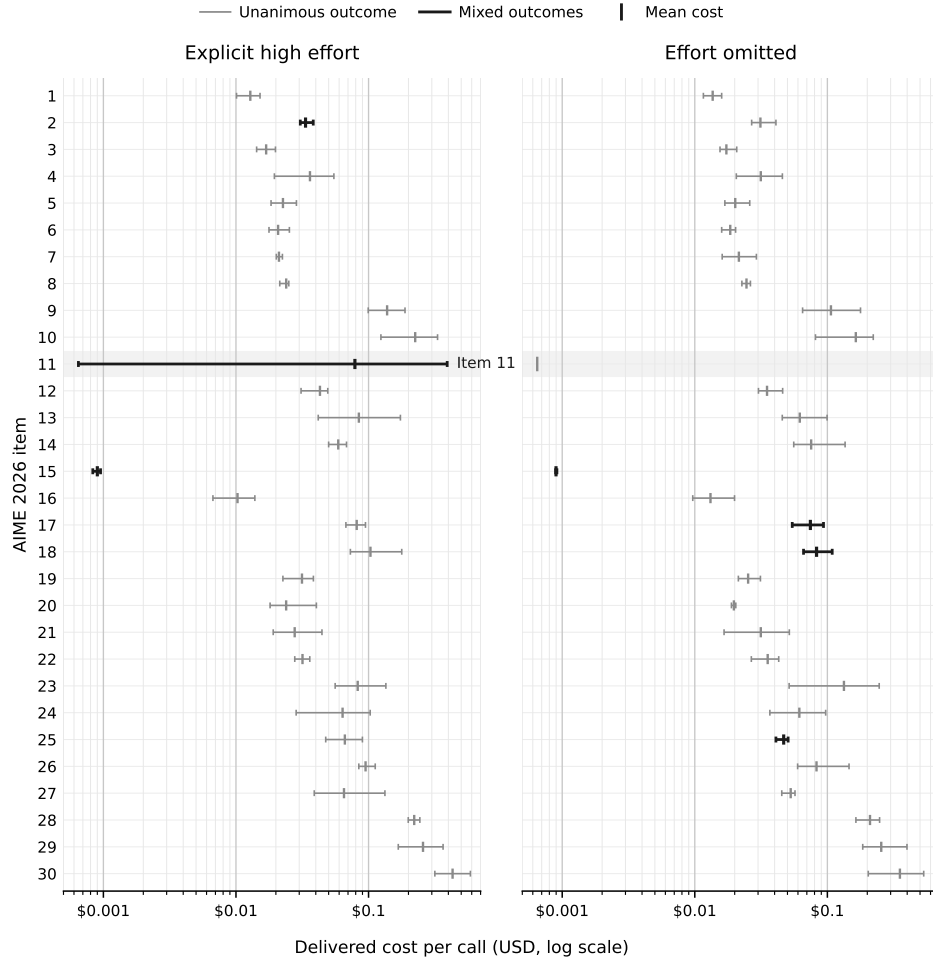


Figure 3. Within-item delivered-cost spans under repeated Sonnet 5 contracts. Each horizontal line spans the minimum to maximum delivered cost across the five registered calls on one AIME 2026 item; end caps mark those endpoints and the interior tick marks the mean. These spans are descriptive ranges, not uncertainty intervals. Zero-width spans are shown as a single mean marker. Darker, heavier lines identify items that produced more than one terminal-outcome category across the five repeats; gray lines identify unanimous outcomes. Sonnet omitted had four mixed-outcome items, compared with three under Sonnet high. Endpoint values are printed at their stored six-decimal precision: on item 11, Sonnet high ranged from \$0.000648 to \$0.391888, while Sonnet omitted produced five identical wrong answers at \$0.000648. Item 11 was selected after outcome inspection as a descriptive illustration; all 30 items remain visible. Costs are shown on a logarithmic scale. Latency is not represented.

inference cost, fixed-query variation, price reversal, or the need for distributional routing, which Chen et al. (2026) report at much greater breadth.

Repeated calls clarify what the two Sonnet contracts delivered on particular items, without establishing how prevalent those behaviors are. In the preliminary item 15 case, one fixed cell paired cheap wrong answers with censored full-rail no-answers. In the registered item 11 case, high effort paired cheap wrong answers with one costly correct computation, whereas Sonnet omitted produced five identical cheap wrong answers. These observations establish two item-specific instances of qualitatively different terminal outcomes under fixed contracts; they do not establish a population rate. Item 11 is also not a replication of item 15 because its second branch was a correct completion rather than a rail-censored no-answer. The main grid contained no `no_answer_rail` outcome and no call at or above 99% rail utilization, so the preregistered conservative gate fired. The rail result therefore narrows to the preliminary existence case, the registered zero counts, and applicable exact single-pass bounds; the degenerate repeated-cell bootstrap intervals provide no informative upper bound. Rail exhaustion also censors latent demand: the study observes delivered cost and the missing answer, not the additional computation that would have completed the response. Consistent with the registered item-count gates, we make no p99 or CVaR claim.

The observed failure modes also differed in buyer legibility across the exact main-grid cells. Terra’s unsuccessful calls ended in visible wrong answers, while the Sonnet cells included no-text no-answer outcomes in addition to wrong answers. This is a descriptive distinction: it concerns what a buyer can observe after purchasing these particular dated contracts. It does not establish a provider-wide failure regime, population prevalence, or a causal difference attributable to the providers. Any use of this contrast must remain conditional on the registered task, prompt, rails, service products, prices, and collection date.

Across the four main cells, the descriptive cost-per-correct point estimates placed Terra omitted lowest, followed by Sonnet omitted, Sonnet high, and Fable omitted. This ordering does not imply dominance: no separate cost-per-correct intervals were registered, and the underlying cost and accuracy intervals overlap. The five-item July 9 bridge supplies historical context rather than additional confirmatory evidence. The three bridge-linked cells shared the same observed accuracy pattern, while Terra’s mean delivered cost lay between the non-contemporaneous GPT-5.4-mini low- and high-effort means. That dated comparison echoes the point-cost decision variable used in the FrugalGPT lineage, but its five-item denominator and non-contemporaneous collection preclude equivalence, tail, latency, or population claims.

Two supporting features make the contract comparison auditable without enlarging its scope. First, the July 2026 model menu assigned different documented and raw-verified meanings to omission and thinking control, including variation within one provider. Omission must therefore be recorded at the exact model-contract level rather than treated as a provider-wide default. Evidence grades remain asymmetric: positive thinking structure verifies realized thinking, while its absence on an easy response cannot distinguish disabled thinking from adaptive thinking that selected zero tokens. Second, the parser, outcome taxonomy, spending ceiling, manifest, and numerical pipeline were frozen before outcome inspection, with later corrections recorded as dated derived artifacts. This governance supports the credibility and auditability of the estimates; it does not expand their empirical reach or restore a first-discovery claim about cost stochasticity or distributions.

Future work should treat effort choice and model choice as dimensions of one registered measurement program. A wider design could cross an effort ladder within a model with a contemporaneous model ladder from the same provider menu, allowing the delivered cost of raising effort to be compared

with the delivered cost of switching models. Those contract-level measurements could also enter effort-aware cascade policies, alongside a deployable confidence or stopping signal, so routing selects among configured contracts rather than model names alone. Because prices, model menus, and omission semantics change, such comparisons would require remeasurement when the available menu changes.

Limitations

The AIME 2025-to-2026 within-index comparison is a freshness diagnostic, not proof that AIME 2026 was uncontaminated. The instant-correct pattern observed at matched 2025 indices disappeared in the 2026 preliminary rows, but the approximately one-month gap from Anthropic’s stated cutoff was narrow. Cutoff metadata, possible benchmark leakage, and later model updates remain uncertain. The main cohort also contains only 30 curated items from one competition-mathematics task family. It is not a probability sample from a formally defined population of model uses. Interpretation beyond the fixed cohort therefore assumes exchangeability with a notional population of comparable contest problems, and should not extend to other tasks without new evidence. Observed accuracy was high in all four main cells, leaving limited headroom for detecting gains; nevertheless, the registered accuracy interval permits an improvement of up to 4.67 percentage points under the explicit-high contract.

The repetition design is asymmetric. Sonnet high and Sonnet omitted were repeated five times per item, but Terra omitted and Fable omitted were observed once per item. The latter cells support cross-item contract summaries but cannot identify within-item repeat variability or conditional outcome composition. Thirty independent items also limit tail resolution. The registered plan permits the median, p90, p95, and maximum, but refuses to estimate p99 with fewer than 100 items or 90%-CVaR with fewer than 50. Repeated calls do not increase those independent-item counts, and numerical computability does not make the ineligible statistics informative.

Latency has a narrower scope than the other call-level measurements. Its reported summaries apply only to the first 20 item-distinct sequential calls in each main cell, not to the full grid or the later bounded-concurrency phase. They are descriptive and do not identify a causal effect of effort on response time. Because the Sonnet cells were dispatched in a fixed high-before-omitted order rather than randomized or counterbalanced, the design also cannot separate request-form differences from systematic dispatch-position or short-run service effects. We therefore interpret the registered contrast as descriptive of the July 18 session, not as a general causal effect of explicitly setting high. More broadly, every main-cell estimate is conditional on that session, the served models and service products observed then, and the price schedule in force on that date. Later model updates, routing changes, prices, tiers, or provider policies could define different contracts and are outside this study.

The GPT-5.4-mini bridge is a five-item, non-contemporaneous comparison collected on July 9. It provides a dated descriptive anchor but supports no equivalence, population accuracy, or cost-and-latency-tail inference. Finally, billed hidden computation is behaviorally measurable but mechanistically opaque: provider records reveal its volume and price, while its content remains unavailable. We can relate delivered cost to the requested contract, visible output, stop state, and terminal outcome, but cannot attribute the primary cost contrast to particular reasoning content or provide the mechanism decomposition that Chen et al. (2026) report.

Data, code, and spend disclosure

AIME 2026 was loaded from `MathArena/aime_2026` (MathArena, 2026) at pinned Hub revision `d2de22f3c656b4f56cf8981212186377d1e23bc3` and is licensed CC BY-NC-SA 4.0. The study release contains item identifiers, contract configurations, row-level measurements, and derived terminal outcomes, but no AIME problem text or reference answers. HEADLINES (Sinha and Khandait, 2020) is licensed CC BY-NC-ND 4.0 and is not redistributed. The numerical release consists of `release_rows.jsonl`, `cell_statistics.json`, `paired_contrasts.json`, `within_item_variation.json`, and `bridge_statistics.json`; `artifact_hashes.json` supplies their SHA-256 integrity pins. The release also includes `plot_figure1_cost_distribution.py`, `plot_figure2_frontier.py`, and `plot_figure3_within_item_span.py`. Figure renders are reproducible from those pinned artifacts but are not shipped. Raw run artifacts remain immutable; corrections and reconciliations are stored as dated derived sidecars rather than rewriting the source rows.

Total API expenditure through the July 18 main collection was \$33.043144: \$31.755307 on Anthropic and \$1.287837 on OpenAI. The total comprises \$3.303106 in precollection runs, \$0.020623 in the July 16 contract probes, \$29.711815 in the main session, and a \$0.007600 Anthropic charge from an interrupted non-streaming request. The supplemental charge is dashboard-confirmed in `results/spend_reconciliation_20260722.json`. Main-session row totals matched both provider dashboards at cent granularity. The probe-day subtotal matches the immutable usage-priced rows, but authenticated dashboard access was unavailable for an aggregate probe-day match. Exact aggregate dashboard evidence is likewise unavailable for the precollection period; its accounting consists of the row-level reconstruction plus the individually confirmed interrupted-request charge. Fifty-four error rows retain null rather than zero cost: 50 belong to one July 7 connection-failed smoke batch, and four are intentional validation-error probes. No usage-priced cost was recorded for those rows; treating the rejected or failed requests as unbilled is documentation-derived rather than request-level billing verification.

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